UNIT II : 1-D Analysis of Trusses and Beams

2.1 Trusses : introduction

A **truss** is a structure of straight slender members joined at their ends by frictionless pins. Loads act only at the joints, so every member carries **axial force only** (tension or compression) - no bending.

- Planar truss : every joint has **2 dof** (u, v) in global X-Y. A bar element has 2 nodes ⇒ **4 dof**.
- Difference from 1-D bar : members are **inclined** at different angles, so the element matrix must be **transformed** from local axis (along the member) to global axes.
- Assumptions : pin joints, loads at joints only, linear elastic material, small displacement, weight of members neglected.

2.2 Stiffness matrix of a 1-D truss element

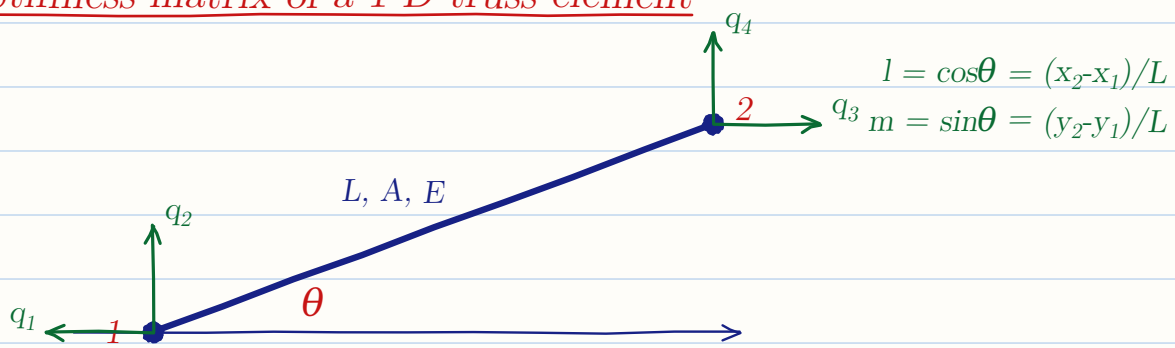


Fig 2.1 Truss element in global co-ordinates

In **local** co-ordinates (x' along the member) the element has 2 dof q'_1, q'_2 and the bar stiffness :

$$[k'] = AE/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Relation between local and global displacements : $u'_1 = q_1 \cos\theta + q_2 \sin\theta = l\, q_1 + m\, q_2$, and $u'_2 = l\, q_3 + m\, q_4$. In matrix form :

$$\{q'\} = [L]\{q\} = \begin{bmatrix} l & m & 0 & 0 \\ 0 & 0 & l & m \end{bmatrix}$$

Strain energy is invariant : $\frac{1}{2}\{q'\}^T[k']\{q'\} = \frac{1}{2}\{q\}^T[k]\{q\} \Rightarrow \mathbf{[k] = [L]^T[k'][L]}$.

Carrying out the multiplication :

$$[k] = AE/L \begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix}$$

$l = \cos\theta = (x_2 - x_1)/L \; m = \sin\theta = (y_2 - y_1)/L \; L = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Remarks: *[k] is symmetric and singular; each row sums to zero; for a horizontal member ($\theta = 0$) it reduces to the 1-D bar matrix. Dof numbering for a node i : (2i−1, 2i). Angle θ is measured counter-clockwise from +X to the direction node 1 → node 2.*

Stress calculation

Stress in the member $\sigma = E\,\varepsilon = E\,(u'_2 - u'_1)/L$:

$$\sigma = (E/L)\ [\ -l\ -m\ l\ m\ ]\ \{q_1\ q_2\ q_3\ q_4\}^T$$

Positive σ = tension, negative = compression. Member axial force $F = \sigma\ A$. Support reactions are obtained from $\{R\} = [K]\{Q\} - \{F\}$, using the rows of the restrained dof.

Procedure for truss analysis

1. Number nodes & elements; write coordinates, A, E of each member.
2. Compute L, l, m for every element; form element [k] (4×4).
3. Assemble global [K] (2 × number of nodes) using dof (2i−1, 2i) of each node.
4. Apply BC (elimination / penalty), solve $[K]\{Q\} = \{F\}$.
5. Find stresses $\sigma = (E/L)[-l\ -m\ l\ m]\{q\}$; reactions from the restrained rows.

2.3 Solved Problem 1 : three-bar truss

Triangular truss : nodes 1 (0,0), 2 (800,0), 3 (400,300) mm. Members (1): 1-2, (2): 2-3, (3): 1-3, all $A = 500\ mm^2$, $E = 2\times10^5\ N/mm^2$. Node 1 pinned, node 2 on a roller ($v_2 = 0$). Vertical load $P = 10\ kN$ (downward) at node 3. Find displacements, stresses, reactions.

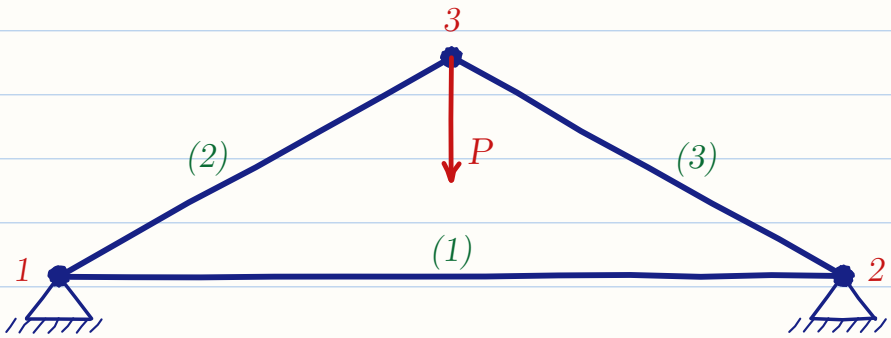


Fig 2.2 (members 1-2 bottom, 2-3 right, 1-3 left)

Step 1 : element data *($AE = 10^8$ N)*

Elem	Nodes	L (mm)	l	m	AE/L (N/mm)
1	1-2	800	1.00	0.00	125000
2	2-3	500	-0.80	0.60	200000
3	1-3	500	0.80	0.60	200000

Step 2 : assemble. *Global dof order : Q_1=u_1, Q_2=v_1, Q_3=u_2, Q_4=v_2, Q_5=u_3, Q_6=v_3. Element dof : (1) = (1,2,3,4), (2) = (3,4,5,6), (3) = (1,2,5,6). Boundary conditions : $Q_1 = Q_2 = Q_4 = 0$, so keep rows/columns 3, 5, 6. Loads : $F_3 = 0$, $F_5 = 0$, $F_6 = -10\,000$ N.*

$$\begin{bmatrix} 253000 & -128000 & 96000 \\ -128000 & 256000 & 0 \\ 96000 & 0 & 144000 \end{bmatrix} \begin{Bmatrix} Q_3 \\ Q_5 \\ Q_6 \end{Bmatrix} = [0,\ 0,\ -10000]^T$$

Step 3 : solve *: $Q_3 = u_2 = 0.0533$ mm, $Q_5 = u_3 = 0.0267$ mm, $Q_6 = v_3 = -0.1050$ mm.*

Step 4 : stresses σ = (E/L)[−l −m l m]{q} :

- *Element 1 (1-2) : σ = 13.33 N/mm^2 (tension), force = 6667 N*
- *Element 2 (2-3) : σ = -16.67 N/mm^2 (compression), force = -8333 N*
- *Element 3 (1-3) : σ = -16.67 N/mm^2 (compression), force = -8333 N*

Step 5 : reactions *: $R_{1x} = 0$ N, $R_{1y} = 5000$ N, $R_{2y} = 5000$ N. Check $\Sigma F_y = 5000 + 5000 = 10\,000$ N, $\Sigma F_x = 0 \approx 0$ - equilibrium satisfied.*

2.4 Solved Problem 2 : two-bar bracket

Two bars hold a load $P = 10$ kN at joint 3. Nodes 1 (0,300), 2 (0,−300) (walls, fixed), 3 (400,0). $A = 500$ mm^2, $E = 2\times10^5$ N/mm^2. Only joint 3 can move : dof Q_5 (u_3), Q_6 (v_3).

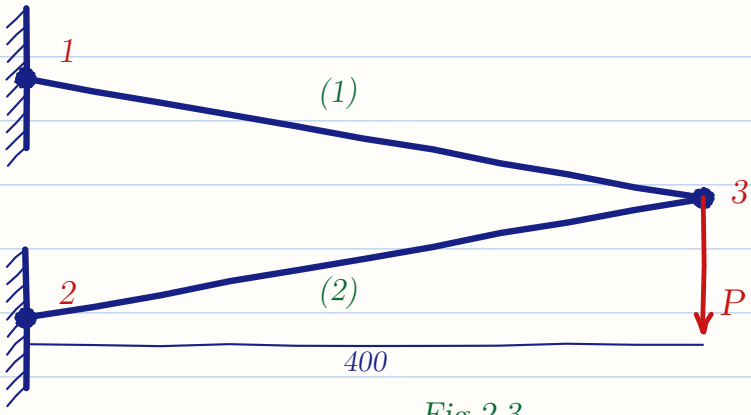


Fig 2.3

Elem	Nodes	L (mm)	l	m	AE/L (N/mm)
1	1-3	500	0.80	-0.60	200000
2	2-3	500	0.80	0.60	200000

For each element only the lower-right 2×2 block (node 3) is needed : $k_{ij} = (AE/L) \times [l^2\ lm\ ;\ lm\ m^2]$.

$$\begin{bmatrix} 256000 & 0 \\ 0 & 144000 \end{bmatrix} \begin{Bmatrix} u_3 \\ v_3 \end{Bmatrix} = [0,\ -10000]^T$$

Result : $u_3 = 0.0000$ mm, $v_3 = -0.0694$ mm. Stresses : $\sigma_1 = 16.67$ N/mm^2, $\sigma_2 = -16.67$ N/mm^2. (Member 1, the upper bar, is in tension; member 2, the lower bar, in compression - matches statics : $F = P/(2\sin\alpha) = 8333$ N.)

2.5 Analysis of Beams

A **beam** carries transverse loads and resists them by bending. Each node has **2 dof** : transverse deflection v and slope (rotation) $\theta = dv/dx$. A 2-noded beam element therefore has **4 dof**.

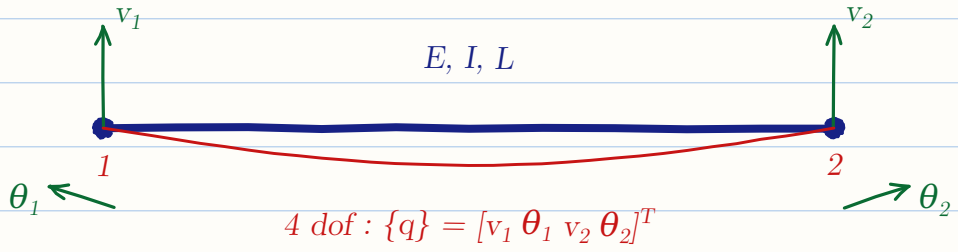


Fig 2.4 Beam element

Beam theory (Euler-Bernoulli) : $M = EI\, d^2v/dx^2$, $V = dM/dx$, $EI\, d^4v/dx^4 = q(x)$. Strain energy $U = \frac{1}{2} \int EI\, (d^2v/dx^2)^2\, dx$. Since second derivatives appear, the shape functions must be **C^1 continuous** (both v and dv/dx continuous between elements) ⇒ **Hermite cubic** polynomials.

Hermite shape functions

Four dof ⇒ cubic : $v = a_0 + a_1\xi + a_2\xi^2 + a_3\xi^3$ with $-1 \le \xi \le 1$, $x = (x_1+x_2)/2 + (L/2)\xi$ and $dv/dx = (2/L)\, dv/d\xi$. Imposing $v(-1) = v_1$, $v'(-1) = \theta_1$, $v(1) = v_2$, $v'(1) = \theta_2$ gives :

$$H_1 = (1/4)(1-\xi)^2(2+\xi) \quad H_2 = (L/8)(1-\xi)^2(1+\xi)$$

$$H_3 = (1/4)(1+\xi)^2(2-\xi) \quad H_4 = (L/8)(1+\xi)^2(\xi-1)$$

$$v(\xi) = H_1v_1 + H_2\theta_1 + H_3v_2 + H_4\theta_2 = [H]\{q\}$$

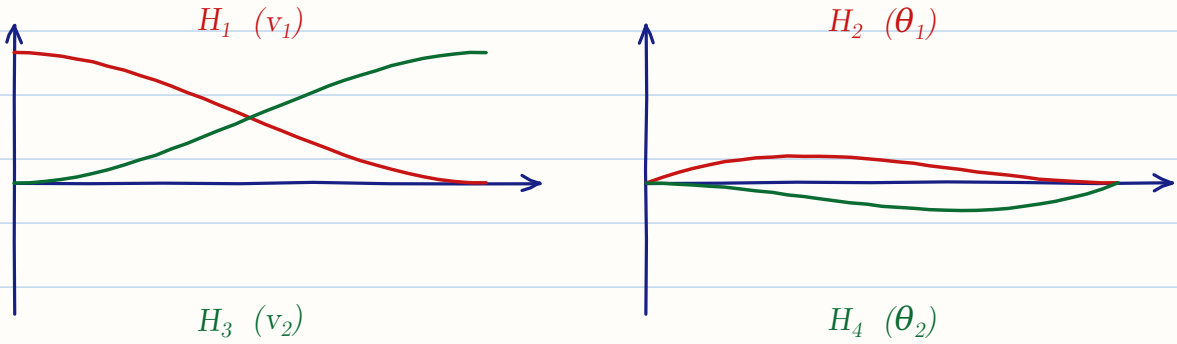


Fig 2.5 Hermite shape functions

Properties : H_1=1 at node 1 and H_1=H_1'=0 at node 2 (similarly H_3); H_2'=1 at node 1 with zero value and slope elsewhere (similarly H_4); $H_1+H_3 = 1$ (rigid-body translation).

Element stiffness matrix

Curvature $d^2v/dx^2 = (4/L^2)\, d^2v/d\xi^2 = [B]\{q\}$, $[B] = (1/L^2)\,[\,6\xi\,,\, L(3\xi-1)\,,\, -6\xi\,,\, L(3\xi+1)\,]$.

$$[k] = \int_0^L [B]^T\, EI\, [B]\, dx = (EI\, L/2) \int_{-1}^{1} [B]^T [B]\, d\xi$$

$$[k] = EI/L^3 \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

DOF order $\{v_1, \theta_1, v_2, \theta_2\}$. $[k]$ is symmetric, singular (2 rigid-body modes : translation and rotation).

Load vector (consistent nodal loads)

$\{f\} = \int [H]^T\, q\, dx$. For a **uniformly distributed load** q (positive in the $+v$ direction) :

$$\{f\} = qL/2 \begin{bmatrix} 1 \\ L/6 \\ 1 \\ -L/6 \end{bmatrix}$$

i.e. end forces $qL/2$ and fixed-end moments $\pm\, qL^2/12$ (same as fixed-end reactions). For a **point load** P at a node, it is added directly to the v dof; an applied couple M_0 to the θ dof. For a point load inside an element, use fixed-end values $P/2$, $PL/8$, $P/2$, $-PL/8$ (load at the middle).

After solving : bending moment at any section $M = EI\, [B]\{q\}$; at node 1 ($\xi=-1$) and node 2 ($\xi=+1$).

2.6 Solved Problems on Beams

Problem 3 : cantilever with end load

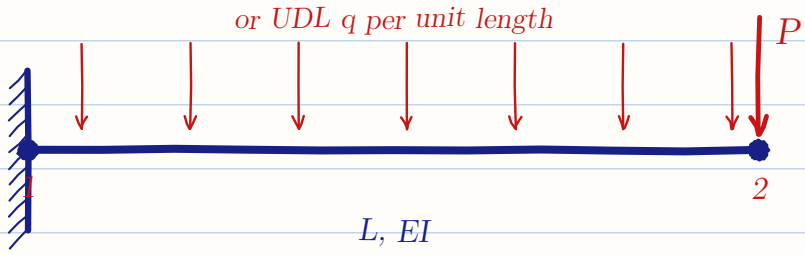


Fig 2.6

One element, $L = 1000$ mm, $E = 2\times10^5$ N/mm^2, $I = 10^7$ mm^4 ($EI = 2\times10^{12}$ N · mm^2), $P = 1000$ N downward at the free end. BC : $v_1 = \theta_1 = 0$. Retain rows 3, 4 :

$$EI/L^3 \begin{bmatrix} 12 & -6L \\ -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} v_2 \\ \theta_2 \end{Bmatrix} = [-P\ ,\ 0]^T$$

Solving : $v_2 = -PL^3/(3EI) = -1000\times10^9/(3\times2\times10^{12}) = -0.1667$ mm (downward), $\theta_2 = -PL^2/(2EI) = -0.00025$ rad. These agree with the **exact** beam-theory results. Reactions from rows 1-2 : $R_1 = +P = 1000$ N upward and $M_1 = PL = 10^6$ N · mm (anticlockwise).

Problem 4 : cantilever with UDL (one element)

Load vector at node 2 : $F = [-qL/2,\ +qL^2/12]^T$ (downward q). Solve EI/L^3 [12 $-6L$; $-6L$ $4L^2$]$\{v_2, \theta_2\} = F$:

$$v_2 = -qL^4/(8EI)\ \theta_2 = -qL^3/(6EI) \text{ (both exact)}$$

Problem 5 : simply supported beam with central load (2 elements)

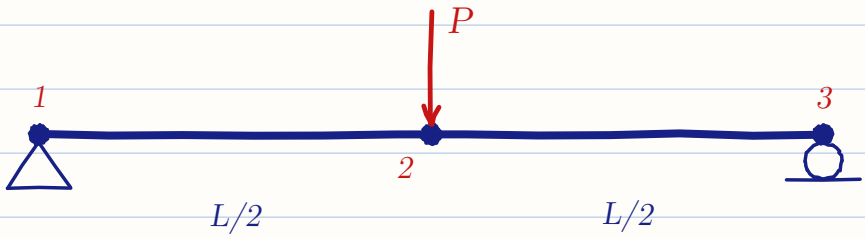


Fig 2.7

Two elements of length $a = L/2$. Nodes 1, 2, 3. BC : $v_1 = v_3 = 0$. Free dof : θ_1, v_2, θ_2, θ_3. Load P downward at node 2 : $F_{v2} = -P$. By symmetry $\theta_2 = 0$ and $\theta_3 = -\theta_1$. The reduced equations give :

$$v_2 = -P(2a)^3/(48EI) = -PL^3/(48EI)\ \theta_1 = -PL^2/(16EI) \text{ (exact)}$$

Moment at the centre $M = EI\,v'' = PL/4$ (obtained from [B] at $\xi = -1$ of element 2). Because the nodal values are exact for beams loaded at nodes, only the interior curve (cubic) is approximate for distributed loads.

Summary: Beam & truss formulas to remember : truss $[k] = AE/L\ [l^2\ lm\ ...]$ $\sigma = (E/L)[-l\ -m\ l\ m]\{q\}$. beam $[k] = EI/L^3\ [12\ 6L\ ...]$, UDL load vector $qL/2\ [1,\ L/6,\ 1,\ -L/6]$, Hermite cubics H_1 to H_4 (C^1 continuity).

2.7 Additional Worked Problems

Direction cosines : quick table

θ (deg)	0	30	45	60	90	120
$l = \cos\theta$	1	0.866	0.707	0.5	0	−0.5
$m = \sin\theta$	0	0.5	0.707	0.866	1	0.866

Always find θ from the coordinates (node 1 → node 2), not from the picture. Swapping node numbers changes the signs of l and m but gives the same $[k]$ and the same stress.

Problem 5 (numerical) : simply supported beam, P = 10 kN at centre

$L = 2000$ mm (two elements of $a = 1000$ mm), $EI = 2\times10^{12}$ N · mm^2. Free dof : $\theta_1,\ v_2,\ \theta_2,\ \theta_3$. Solution of the reduced 4×4 system :

θ_1 (rad)	v_2 (mm)	θ_2 (rad)	θ_3 (rad)
-0.001250	-0.8333	0.000000	0.001250

Exact : $v_2 = -PL^3/48EI = -10^4\times8\times10^9/(48\times2\times10^{12}) = -0.8333$ mm ; $\theta_1 = -PL^2/16EI = -0.00125$ rad. Reactions : $R_1 = R_3 = P/2 = 5000$ N. Maximum bending moment $PL/4 = 5\times10^6$ N · mm.

Problem 6 : three-bar truss meeting at one joint

Three bars ($A = 500$ mm^2, $E = 2\times10^5$ N/mm^2) join at node 4 (400,0). Their other ends are fixed at 1 (0,300), 2 (0,−300), 3 (0,0). Joint 4 carries 5 kN horizontally (+X) and 10 kN downward. Only dof $Q_7 = u_4$, $Q_8 = v_4$ are free.

Elem	Nodes	L	l	m	AE/L
1	1-4	500	0.80	-0.60	200000
2	2-4	500	0.80	0.60	200000
3	3-4	400	1.00	0.00	250000

$$\begin{bmatrix} 506000 & 0 \\ 0 & 144000 \end{bmatrix} \begin{Bmatrix} u_4 \\ v_4 \end{Bmatrix} = [5000,\ -10000]^T$$

Solution : u_4 = 0.0099 mm, v_4 = -0.0694 mm. Stresses : σ_1 = 19.83, σ_2 = -13.50, σ_3 = 4.94 N/mm^2. The horizontal bar 3 takes the horizontal load; bars 1 and 2 take the vertical load (statically indeterminate structure - FEM handles it without extra work).

Beam [B] matrix and a typical stiffness term

Differentiate the Hermite functions twice with respect to ξ and multiply by $(4/L^2)$:

	H_1	H_2	H_3	H_4
d^2H/dx^2	$6\xi/L^2$	$(3\xi-1)/L$	$-6\xi/L^2$	$(3\xi+1)/L$

Example : $k_{11} = EI\ (L/2) \int_{-1}^{1} (6\xi/L^2)^2\ d\xi = EI\ (L/2)(36/L^4)(2/3) = 12EI/L^3$; $k_{22} = EI\ (L/2)(1/L^2) \int (3\xi-1)^2\ d\xi = EI/(2L) \times 8 = 4EI/L$.

Problem 7 : propped cantilever with UDL (1 element)

Fixed at node 1 ($v_1 = \theta_1 = 0$), roller at node 2 ($v_2 = 0$). Only θ_2 is free : $(4EI/L)\ \theta_2 = F_{\theta 2} = +qL^2/12 \Rightarrow \theta_2 = qL^3/(48EI)$. Reaction at the roller $R_2 = (EI/L^3)(-6L\ \theta_2) + qL/2 = 3qL/8$ (exact; the fixed end then carries $5qL/8$).

Problem 8 : simply supported beam with UDL (1 element)

$$EI/L^3 \begin{bmatrix} 4L^2 & 2L^2 \\ 2L^2 & 4L^2 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = [-qL^2/12\ ,\ +qL^2/12]^T$$

$\theta_1 = -qL^3/(24EI)$, $\theta_2 = +qL^3/(24EI)$ (exact end slopes). Mid-span deflection from the interpolation : $v(0) = H_2(0)\theta_1 + H_4(0)\theta_2 = (L/8)(\theta_1 - \theta_2) = -qL^4/(96EI) = -0.0104\ qL^4/EI$, whereas exact $= -5qL^4/(384EI) = -0.0130\ qL^4/EI$. The error ($\approx$ 20 %) is removed by using more elements.

Convergence check : cantilever tip load with 1, 2 elements

Two elements of 500 mm, $EI = 2\times10^{12}$ N · mm^2, P = 1000 N at the tip. Unknowns v_2, θ_2, v_3, θ_3 after assembling two 4×4 matrices overlapping at node 2 : v_2 = -0.0521 mm, θ_2 = -0.000188 rad, v_3 = -0.1667 mm, θ_3 = -0.000250 rad. These coincide with the exact values ($PL^3/3EI$ = 0.1667 mm), because Hermite cubics can represent the exact cubic deflection curve of a tip-loaded cantilever.

Shear force and moment from FE results

- Bending moment in an element : $M(\xi) = EI\,[B]\{q\}$; at node 1 $M_1 = EI/L^2$ $[-6, -2L, 6, -4L]\{q\}$, at node 2 $M_2 = EI/L^2$ $[6, 4L, -6, 2L]\{q\}$.
- Shear force $V = dM/dx = (EI/L^3)$ $[12\ ,\ 6L\ ,\ -12\ ,\ 6L]\{q\}$ (constant in the element since the displacement is cubic).
- Alternatively use the nodal equilibrium : element end forces $\{f\} = [k]\{q\} - \{\text{fixed-end loads}\}$.

Truss vs Bar vs Beam : comparison

Element	Nodes	dof / node	Shape fn	Matrix size	Continuity
1-D bar	2	1 (u)	linear	2 × 2	C^0
Plane truss	2	2 (u,v)	linear	4 × 4	C^0
Beam	2	2 (v,θ)	Hermite cubic	4 × 4	C^1

2.8 Problem 9 : cantilever with UDL, two elements

Cantilever $L = 1000$ mm, $EI = 2\times10^{12}$ N · mm^2, UDL $q = 10$ N/mm downward. Two elements, $L_e = 500$ mm. Element load vector (downward q) : $\{f\} = -qL_e/2\ [1,\ L_e/6,\ 1,\ -L_e/6]^T = [-2500,\ -208333,\ -2500,\ 208333]^T$ (N, N · mm).

Assemble : the shared node 2 receives the sum of the contributions - $F_{v2} = -5000$ N, $F_{\theta 2} = 0$ N · mm (end moments cancel), $F_{v3} = -2500$ N, $F_{\theta 3} = 208333$ N · mm.

Apply $v_1 = \theta_1 = 0$ and solve the 4×4 reduced system : $v_2 = -0.2214$ mm, $\theta_2 = -0.000729$ rad, $v_3 = -0.6250$ mm, $\theta_3 = -0.000833$ rad. Exact tip deflection $-qL^4/(8EI) = -0.6250$ mm : nodal values are exact for a uniformly loaded beam.

Check: Nodal values for beams are exact for any load that is a polynomial of degree ≤ 3 when consistent nodal loads are used; the error lies only in the interpolated curve inside the element. Check the sum of vertical nodal loads = total load (here 10×1000 = 10 000 N).

Problem 10 : numerical element matrix and stress of one inclined bar

Bar from node 1 (0,0) to node 2 (300,400) mm; $A = 500\ mm^2$, $E = 2\times10^5\ N/mm^2$. $L = 500\ mm$, $l = 0.6$, $m = 0.8$, $AE/L = 2\times10^5\ N/mm$.

$$[k] = \begin{bmatrix} 72000 & 96000 & -72000 & -96000 \\ 96000 & 128000 & -96000 & -128000 \\ -72000 & -96000 & 72000 & 96000 \\ -96000 & -128000 & 96000 & 128000 \end{bmatrix}$$

Node 1 fixed, node 2 displaced $u_2 = 0.10\ mm$, $v_2 = 0.05\ mm$. Elongation of the bar $= l\ u_2 + m\ v_2 = 0.06 + 0.04 = 0.10\ mm$. Strain $= 0.10/500 = 2\times10^{-4}$, so $\sigma = E\varepsilon = 40\ N/mm^2$ (tension) and axial force $= 40\times500 = 20\ 000\ N$. Forces at node 2 : $F_x = 12000\ N$, $F_y = 16000\ N$ ($= 20\ 000 \times l$ and $\times\ m$).

Common mistakes to avoid

- Using degrees/radians wrongly in l, m; mixing units (N, mm, MPa); forgetting that dof of node i are (2i−1, 2i).
- Applying the support condition of a **roller** wrongly (only the dof normal to the rolling surface is zero).
- In beams : writing the load vector with wrong signs (check direction of q and sign of end moments).

Formula sheet (Unit II)

Item	Truss element	Beam element
dof / element	4 (u,v at 2 nodes)	4 (v,θ at 2 nodes)
Stiffness	AE/L [l^2, lm, ...]	EI/L^3 [12, 6L, ...]
Strain / curvature	$\varepsilon = (1/L)[-l\ -m\ l\ m]\{q\}$	$\kappa = (1/L^2)[6\xi\ ...]\{q\}$
Stress	$\sigma = E\,\varepsilon$	$\sigma = -E\,y\,\kappa$ or $M\,y\,/\,I$
Load vector	nodal loads only	$qL/2$ [1, L/6, 1, −L/6]

Short answer questions

Q1. Why is a truss element different from a bar element ? - Members are oriented at an angle; each node has 2 dof so a transformation from local to global axes is needed.

Q2. What is the size of the truss element matrix ? - 4×4 (2 nodes × 2 dof) in 2-D; 6×6 in 3-D.

Q3. Why Hermite polynomials for beams ? - Slope continuity (C^1) is necessary because strain energy contains d^2v/dx^2.

Q4. What is the degree of the beam interpolation ? - Cubic (4 dof per element).

Q5. What are consistent nodal loads ? - Loads derived from the same shape functions, $\{f\} = \int [H]^T q \, dx$; equal to fixed-end reactions.

Q6. Sign of stress in truss member ? - positive ⇒ tension, negative ⇒ compression.

Q7. Number of rigid-body modes of a beam element ? - Two (translation and rotation); a plane truss element has three (2 translations + 1 rotation) in the global system.

Expected questions

- Derive the stiffness matrix of a 2-D truss element; write stress expression. Solve a 3-member truss for displacements, stresses and reactions.
- Derive Hermite shape functions for a beam; derive [k] and load vector for UDL. Solve cantilever / simply supported beam using 1-2 elements.
- Why is C^1 continuity required for beam elements but C^0 suffices for bars?